

Complexity Theory

Final Exam - Solutions

2022

2ND YEAR - ENSIA

EXERCISES

Exercise 1 (5 points) :

1. Given the language L over the alphabet $\Sigma = \{0, 1\}$ which contains palindromes having a length less than 10,000. Is L regular ? justify your answer. (1 point)

L is finite $\Rightarrow$ Regular

2. Could we find an unambiguous grammar (CFG) for each context-free language? Justify your answer. (1 point)

No, if the language is inherently ambiguous, we cannot find an unambiguous grammar (CFG)

3. Can we conclude that a language L is regular if a selected word "s" with a length greater than the pumping length "p" satisfies the pumping lemma without any contradictions? Justify your answer. (1 point)

No, the pumping lemma is used to prove that the language isn't regular.

Or

No, Maybe another string with length greater than the pumping length leads to a contradiction.

4. Give an example of two context-free languages to show that the intersection is not closed under the set of context-free languages. (1 point)

$L1 = \{0^n 1^n 1^* \mid n \in \mathbb{N}\}$ $L2 = \{0^* 1^n 1^n \mid n \in \mathbb{N}\}$ are CFLs?

$L1 \cap L2 = \{0^n 1^n 1^n \mid n \in \mathbb{N}\}$ which is not a CFL

Any example that satisfies this should be accepted

5. In Turing machines, explain how the stay-put (S) transition $(q_0, 0) \Rightarrow (q_1, 1, S)$ can be implemented with only the conventional transitions {Left (L) , Right (R)} after each action. The tape alphabets are $\Gamma = \{0, 1, \square\}$ (where $\square$ is the blank symbol). (1 point)

$(q_0, 0) \Rightarrow (q_{stay}, 1, R)$

$(q_{stay}, 0) \Rightarrow (q_1, 0, L)$

$(q_{stay}, 1) \Rightarrow (q_1, 1, L)$

$(q_{stay}, \square) \Rightarrow (q_1, \square, L)$

Or

$(q_0, 0) \Rightarrow (q_{stay}, 1, L)$

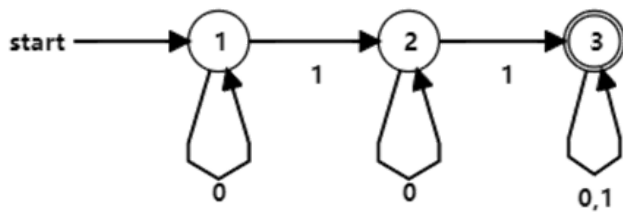
$(q_{stay}, 0) \Rightarrow (q_1, 0, R)$

$(q_{stay}, 1) \Rightarrow (q_1, 1, R)$

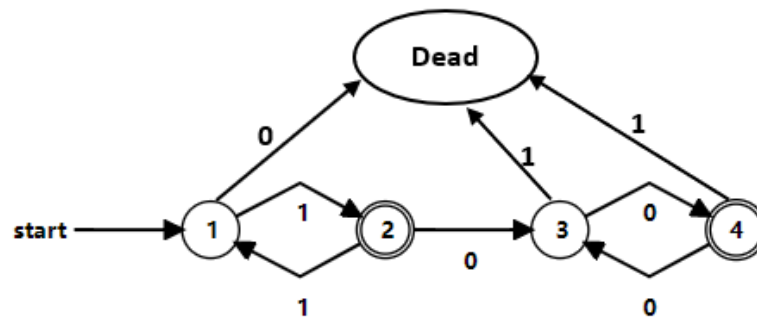
$(q_{stay}, \square) \Rightarrow (q_1, \square, R)$

Exercise 2 (5 points) :

1. Draw a DFA for the language that contains at least two ones. (2 point)



2. Let $D = \{w \mid w \text{ contains an even number of 0's and an odd number of 1's and does not contain the substring } 01\}$. Give the corresponding DFA with five states. (1 points)



3. Let $\Sigma = \{0, 1, +, =\}$ and $ADD = \{x=y+z \mid x, y, z \text{ are binary integers, and } x \text{ is the sum of } y \text{ and } z\}$. Show that ADD is **not** regular. (2 points)

Examples of words from the language ADD is $\{11=10+01, 100=11+1, \dots \text{etc} \}$

Assume that ADD is regular $\Rightarrow$ The pumping lemma is satisfied

Let's take $s = "1^p = 1^{p-1}0 + 1"$: $|s| = 2p + 3 > p$

$s = xyz$ and $|y| > 0$ and $|xy| \leq p \Rightarrow y = 1^l$: $0 < l \leq p$

$xy^2z = "1^{p+l} = 1^{p-1}0 + 1"$ $\Rightarrow$ the equation is incorrect because only the result 1^{p+l} is modified

$\Rightarrow xy^2z \notin ADD$ for $0 < l \leq p$

$\Rightarrow$ The pumping lemma is not satisfied

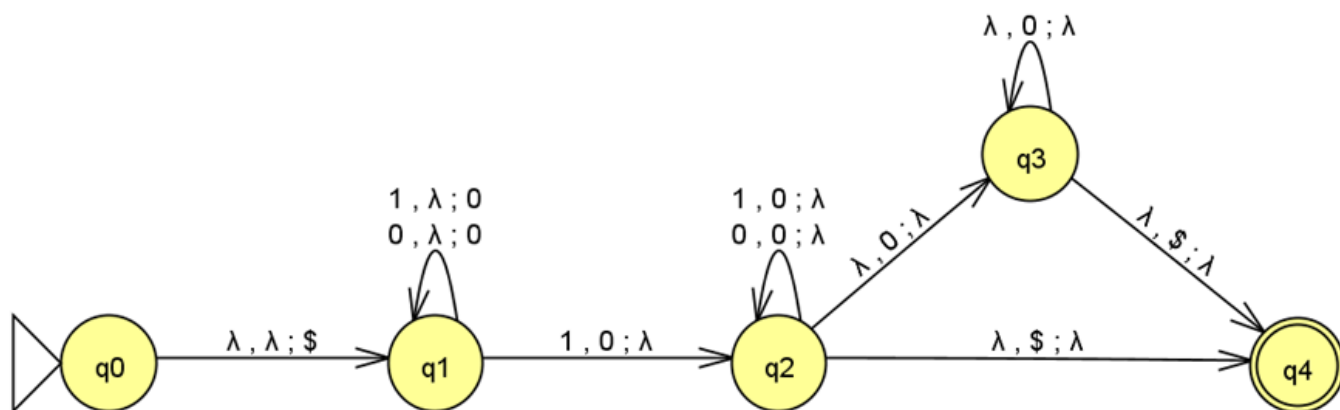
$\Rightarrow$ Contradiction

$\Rightarrow$ ADD is not regular

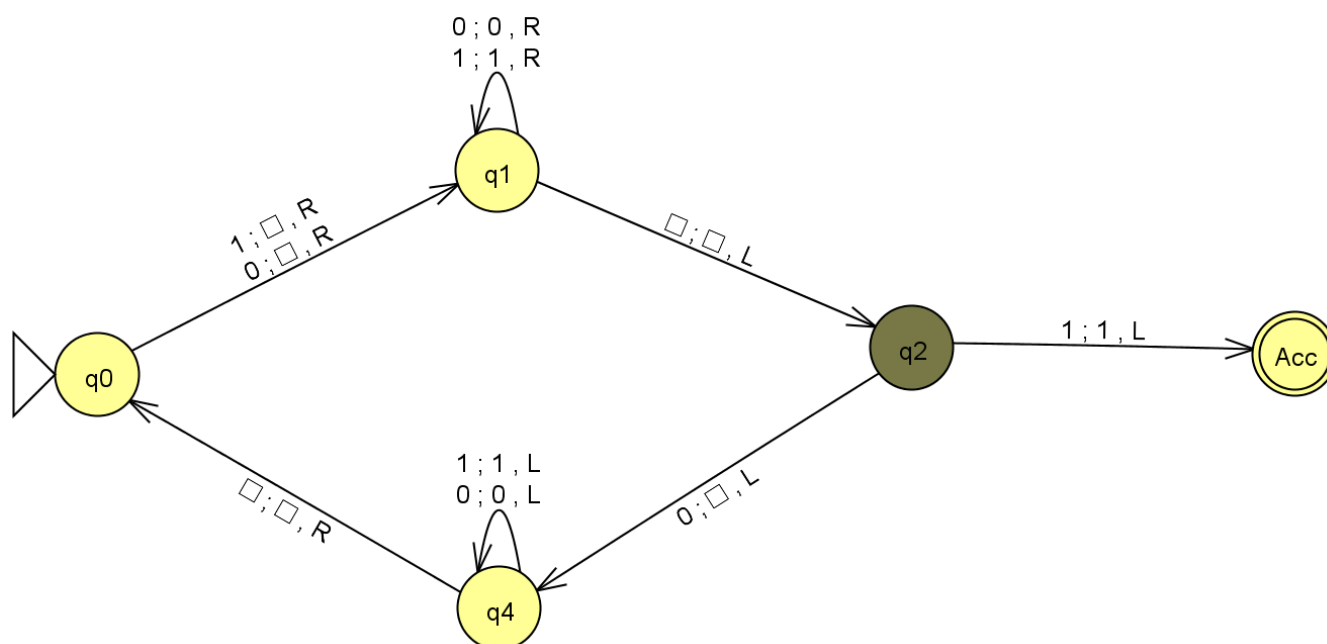
Exercise 3 (5 points) :

Let $\Sigma = \{0, 1\}$ and let B be the collection of strings that contain at least one 1 in their second half. In other words, $B = \{uv \mid u \in \Sigma^*, v \in \Sigma^*1\Sigma^* \text{ and } |u| \geq |v|\}$.

- a. Give a PDA that recognizes B. (2 points)



b. Give a deterministic Turing machine that decides B. (3 points)



Exercise 4 (5 points) :

Given the Turing machine M which converts a binary representation of a number to its equivalent unary representation, and removes all symbols other than the unary representation from the tape.

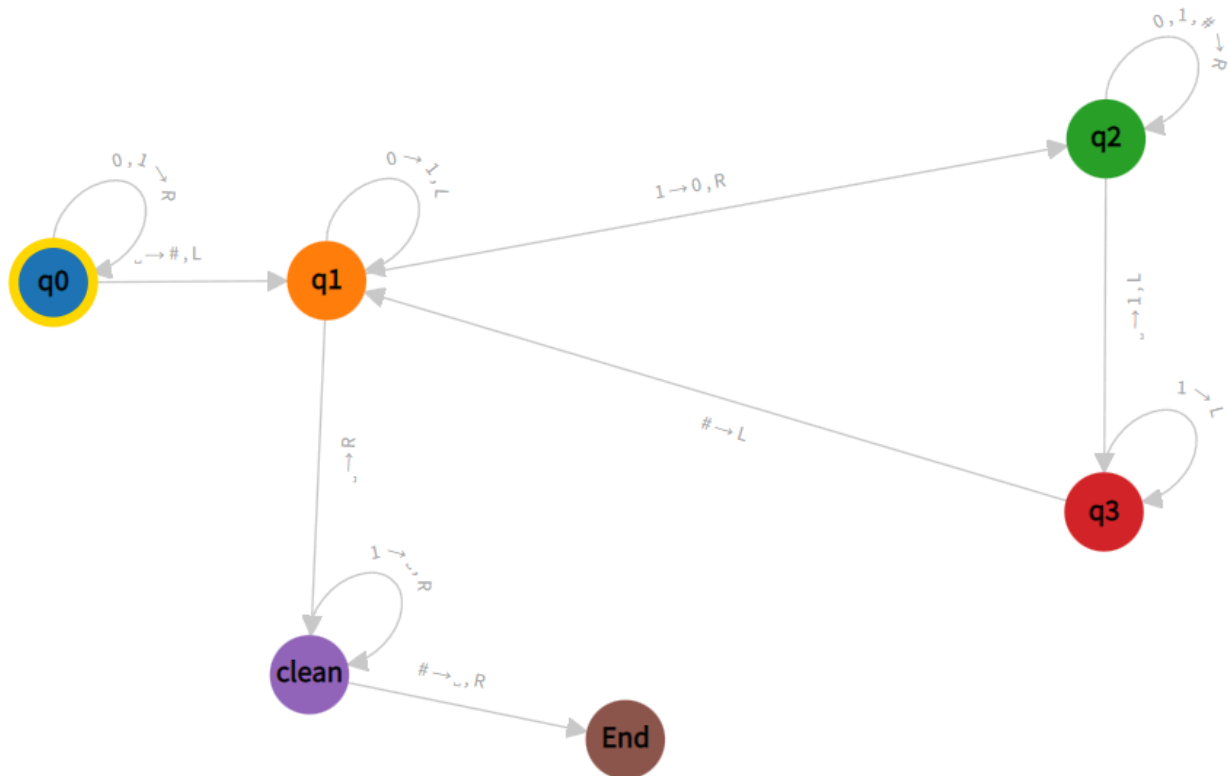
Examples: $0 \Rightarrow \square$, $1 \Rightarrow 1$, $10 \Rightarrow 11$, $101 \Rightarrow 11111$.

1. Write the algorithm for the Turing machine M in clear and precise English (Pseudo-code is accepted). (2 points)
2. Draw the state diagram for the Turing machine M. (2 points)
3. Run a simulation of the Turing machine M on input "10" showing the configuration table with each step as a show highlighting the current step and the position of the tape head. (1 point)

4. Algorithm:

- a. Add separator “#” after the binary number
- b. If the binary number is not zero
 - i. Decrement the binary number
 - ii. Append “1” to the unary representation
- c. Else (the binary number becomes 0)
 - i. Clean the binary number and the separator
 - ii. End of the conversion
- d. Return to (b)

5. Turing machine



6. The simulation

Time	State	Configurations						
0	q0	□	1	0	□	□	□	□
1	q0	□	1	0	□	□	□	□
2	q0	□	1	0	□	□	□	□
3	q1	□	1	0	#	□	□	□
4	q1	□	1	1	#	□	□	□
5	q2	□	0	1	#	□	□	□
6	q2	□	0	1	#	□	□	□
7	q2	□	0	1	#	□	□	□

8	q3	☐	0	1	#	1	☐	☐
9	q1	☐	0	1	#	1	☐	☐
10	q2	☐	0	0	#	1	☐	☐
11	q2	☐	0	0	#	1	☐	☐
12	q2	☐	0	0	#	1	☐	☐
13	q3	☐	0	0	#	1	1	☐
14	q3	☐	0	0	#	1	1	☐
15	q1	☐	0	0	#	1	1	☐
16	q1	☐	0	1	#	1	1	☐
17	q1	☐	1	1	#	1	1	☐
18	clean	☐	1	1	#	1	1	☐
19	clean	☐	☐	1	#	1	1	☐
20	clean	☐	☐	☐	#	1	1	☐
21	accept	☐	☐	☐	☐	1	1	☐